

Class 18: Pertussis and the CMI-PB project

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Background

Pertussis (whooping cough) is a common lung infection caused by the bacteria B. Pertussis. It can infect anyone but is most deadly for infants (under 1 year of age).

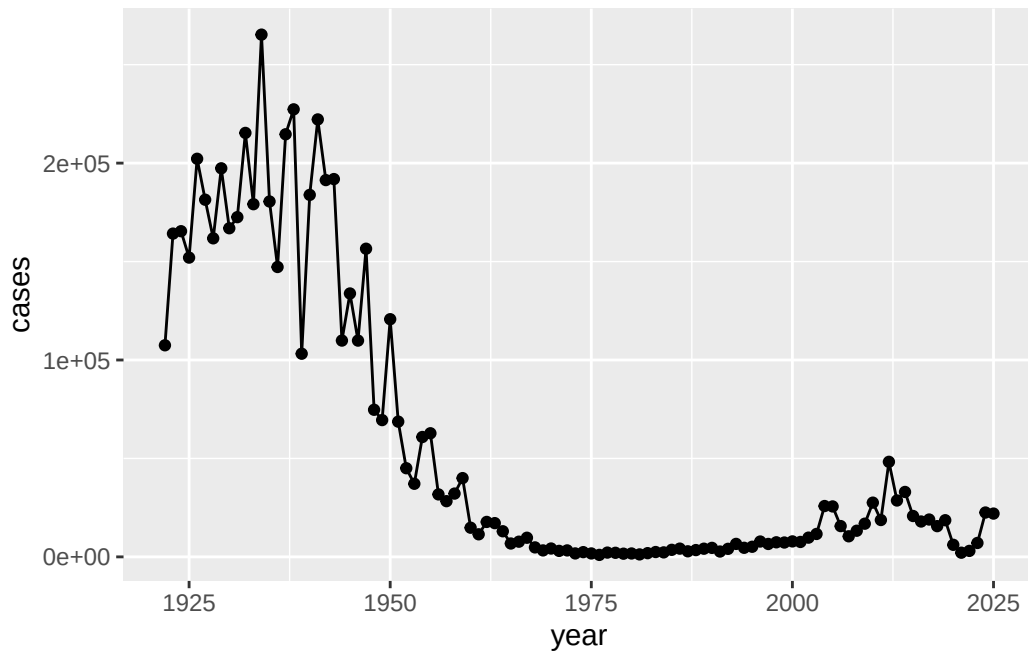
CDC tracking data

The CDC tracks the number of pertussis cases: https://www.cdc.gov/pertussis/php/surveillance/pertussis-cases-by-year.html?CDC_AAref_Val=https://www.cdc.gov/pertussis/surveillance/cases-by-year.html

I want a plot of year vs cases

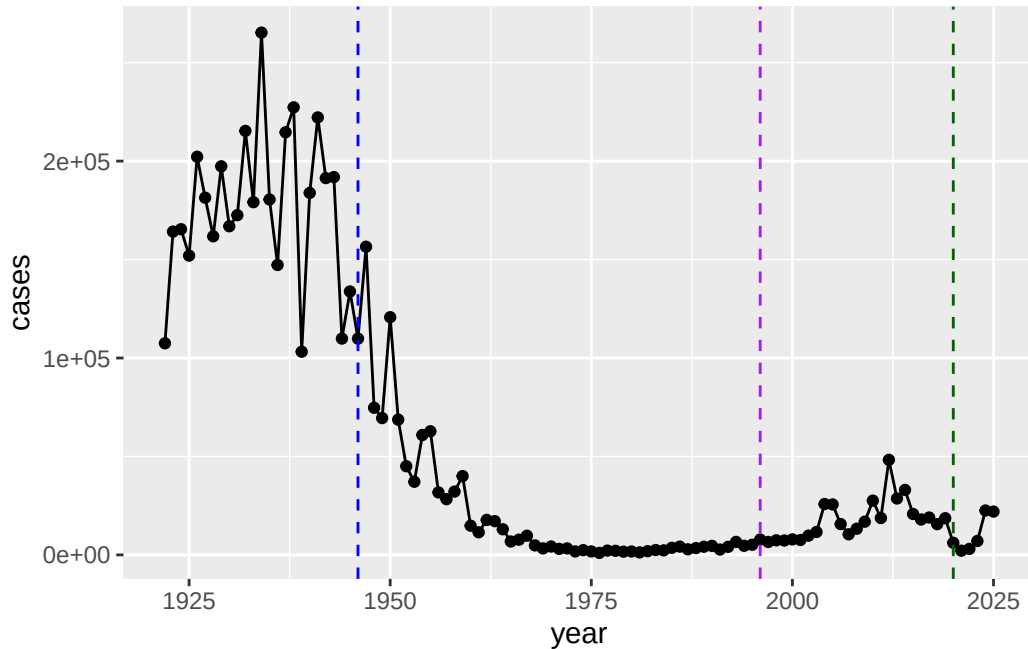
```
library(ggplot2)

ggplot(cdc)+
  aes(year, cases)+
  geom_point() +
  geom_line()
```



Q. Add anotation lines for the major milestones of wP vaccination roll-out (1946) and the switch to the aP vaccine (1996).

```
ggplot(cdc)+
  aes(year, cases)+
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="blue", lty=2)+
  geom_vline(xintercept = 1996, col="purple", lty=2) +
  geom_vline(xintercept = 2020, col="darkgreen", lty=2)
```



After 1946 (wP vaccination roll-out) we see the the cases of Pertussis starts to decrease dramatically. Then in 1946 (switch to the aP vaccine) we start to see an increase in cases. This could be due to a few reasons for this. One could be an increase in people getting testing, as well as more sensitive PCR-based testing. Another reason could be that the aP vaccine has a smaller efficacy when compared to the wP vaccine. This means that after about 12 years the aP vaccine will need a booster shot while the wP vaccine does not.

Exploring CMI-PB data

The CMI-PM project < <https://www.cmi-pb.org/> > mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Basically, make available a large dataset on the immune response to Pertussis. They use a “booster” vaccination as a proxy for Pertussis infection.

They make their data available as JSON format API. We can read this into R with the `read_json()` function from the **jsonlite** package:

```
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many aP and wP individuals are there in this `subject` table?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. How many male/female are there?

```
table(subject$biological_sex)
```

```
Female   Male
  112    60
```

Q. What is the break down of `biological_sex` and `race`?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

Q. Is this representative of the US population?

This data is not representative of the US population. This study was done during the 2020 lockdown and this data is more representative of the UCSD population.

We can read more tables from CMI-PB database

```
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost
1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7
5	5	1	11
6	6	1	32

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

```
head(ab_titer)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

To make sense of all this data we need to “join” (a.k.a. “merge” or “link”) all thses tables together. Only then will you know that a given Ab measurement (from the `ab_titer` table) was collected on a certain date (from the `specimen` table) from a certain wP or aP subject (from the `subject` tabel).

We can is `dplyr` and the `*join()` family of function to do this.

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

Let's do one more `inner_join()` to join the `ab_titer` with all our `meta` data.

```
abdata <- inner_join(ab_titer, meta)
```

Joining with ``by = join_by(specimen_id)``

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	UG/ML	2.096133	1	wP	Female
2	IU/ML	29.170000	1	wP	Female
3	IU/ML	0.530000	1	wP	Female
4	IU/ML	6.205949	1	wP	Female
5	IU/ML	4.679535	1	wP	Female
6	IU/ML	2.816431	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

```

4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                -3                        0          Blood
2                -3                        0          Blood
3                -3                        0          Blood
4                -3                        0          Blood
5                -3                        0          Blood
6                -3                        0          Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1

```

Q. How many different Ab “isotype” values are in this dataset?

```
table(abdata$isotype)
```

```

IgE  IgG  IgG1  IgG2  IgG3  IgG4
6698 7265 11993 12000 12000 12000

```

Q. How many different “antigen” values are in this dataset?

```
table(abdata$antigen)
```

```

  ACT  BETV1    DT  FELD1    FHA  FIM2/3  LOLP1    LOS Measles    OVA
1970  1970  6318  1970  6712  6318  1970  1970  1970  6318
  PD1    PRN    PT   PTM  Total    TT
1970  6712  6712  1970   788  6318

```

Let’s focus on IgG isotype

```

igg <- abdata |>
  filter(isotype=="IgG")
head(igg)

```


	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	IU/ML	0.530000	1	wP	Female
2	IU/ML	6.205949	1	wP	Female
3	IU/ML	4.679535	1	wP	Female
4	IU/ML	0.530000	3	wP	Female
5	IU/ML	6.205949	3	wP	Female
6	IU/ML	4.679535	3	wP	Female

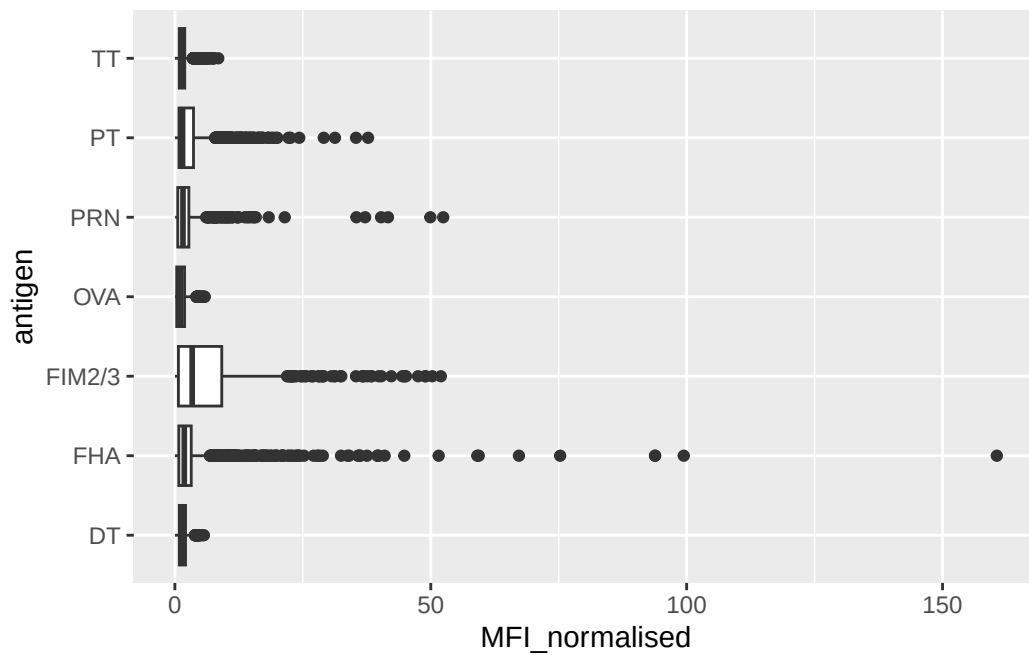
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3		Blood
2	-3		Blood
3	-3		Blood
4	-3		Blood
5	-3		Blood
6	-3		Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

Let's make a plot of MFI_normalised values for all antigenvalues.

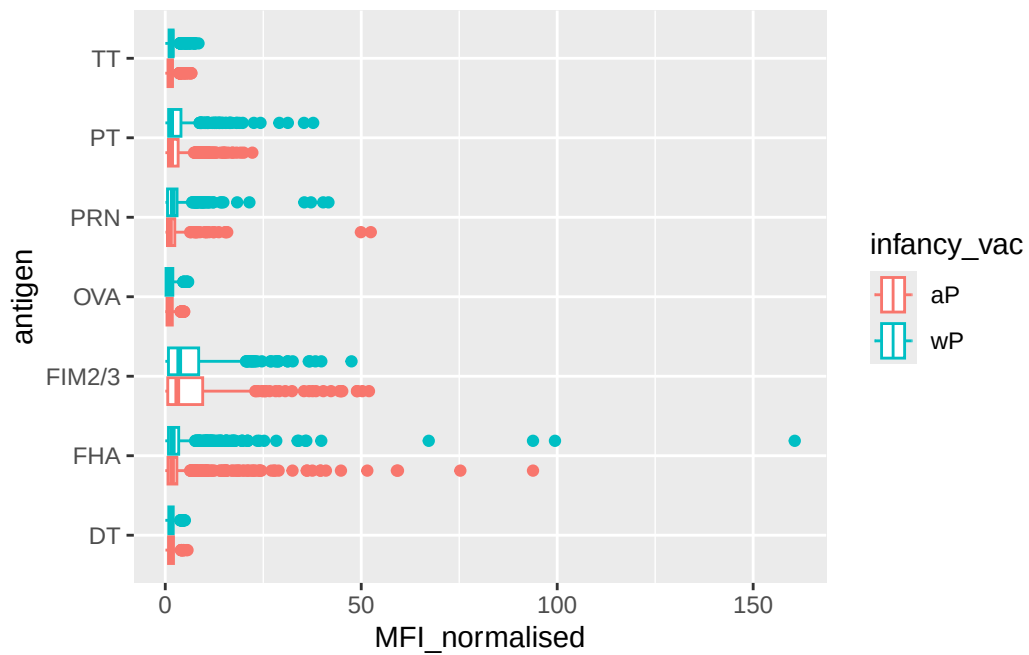
```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot()
```



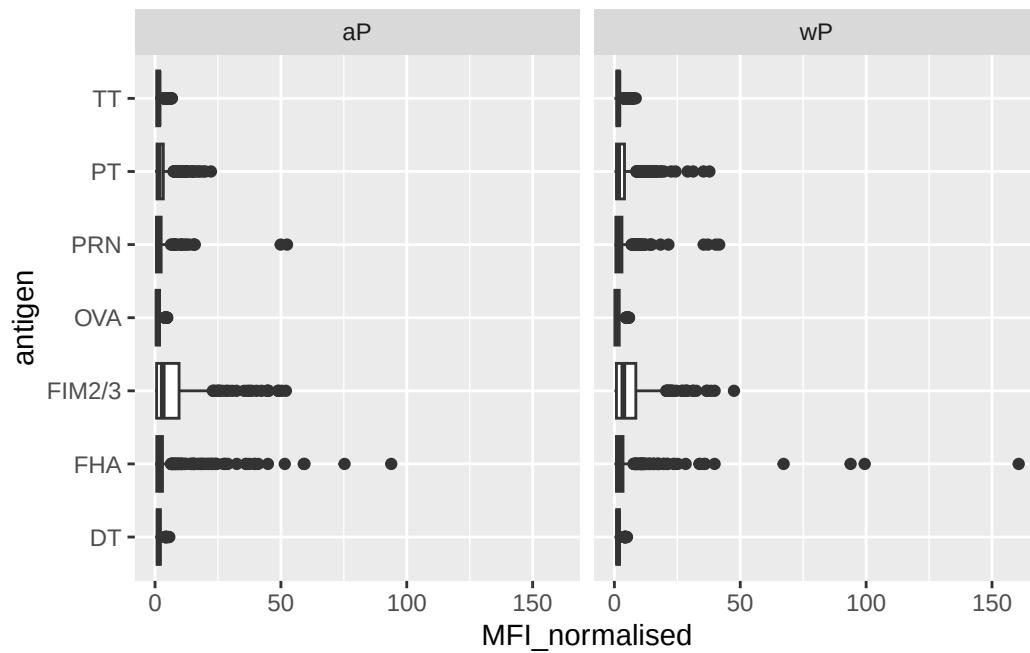
The antigen “PT”, “FIM2/3”, and “FHA” appear to have the widest range of values.

Q. Is there a difference for these responses between aP and wP individuals?

```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot()
```

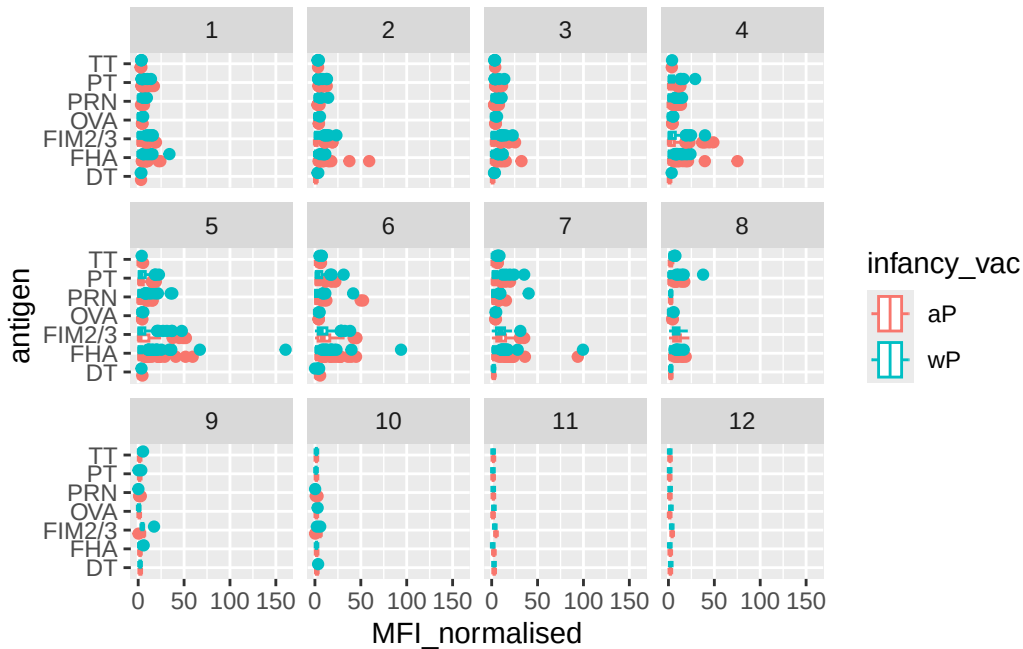


```
ggplot(igg)+
  aes(MFI_normalised, antigen)+
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



Q. Is there a difference with time (i.e before booster shot vs after booster shot)?

```
ggplot(igg)+
  aes(MFI_normalised, antigen, col=infancy_vac)+
  geom_boxplot() +
  facet_wrap(~visit)
```



```
abdata.21 <- abdata |> filter(dataset=="2021_dataset")
abdata.21 |> filter(isotype=="IgG", antigen=="PT") |>
  ggplot()+
  aes(x=planned_day_relative_to_boost, y=MFI_normalised,col=infancy_vac, group=subject_id)+
  geom_point() +
  geom_line() +
  geom_smooth(aes(group = infancy_vac), se =FALSE, linewidth=2, span =0.3) +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
        subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at -0.6

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 3.6

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 0

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
: There are other near singularities as well. 11364
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
: pseudoinverse used at -0.6
```

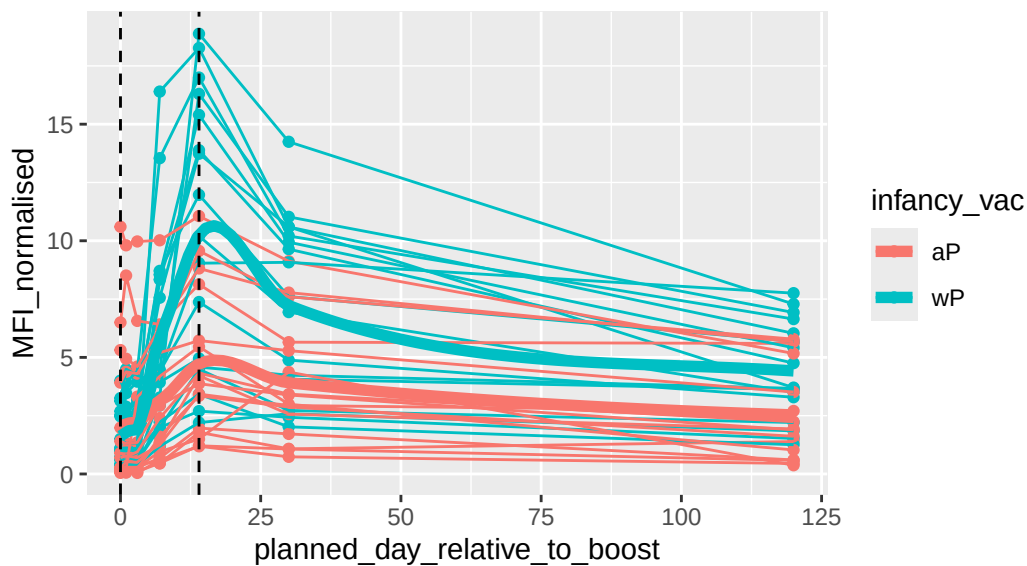
```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
: neighborhood radius 3.6
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
: reciprocal condition number 0
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
: There are other near singularities as well. 11364
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



Obtaining CMI-PB RNASeq data

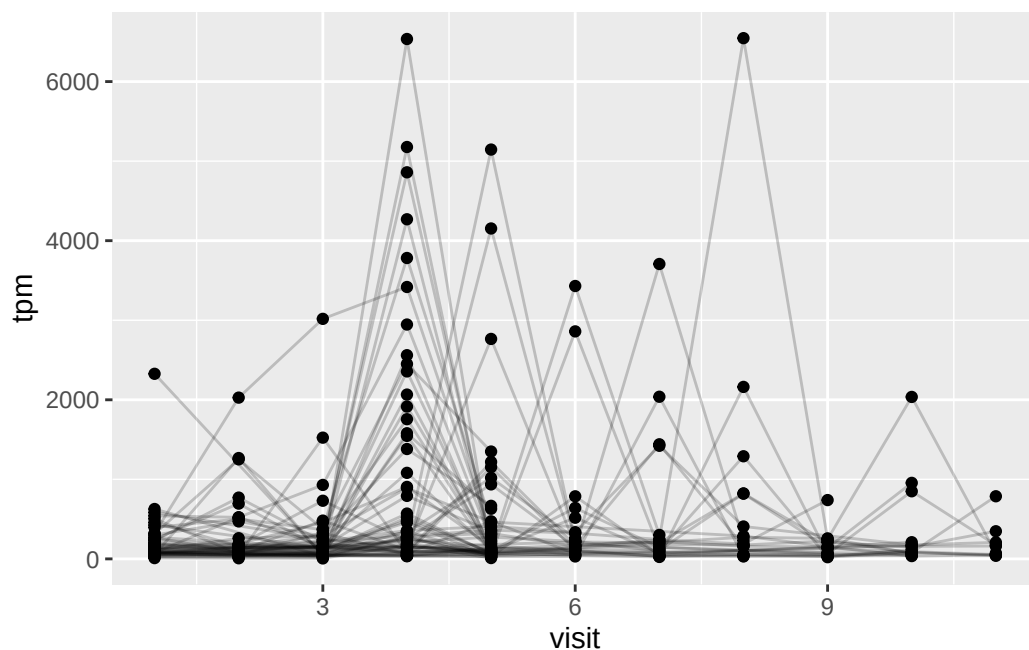
```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"  
rna <- read_json(url, simplifyVector = TRUE)
```

```
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

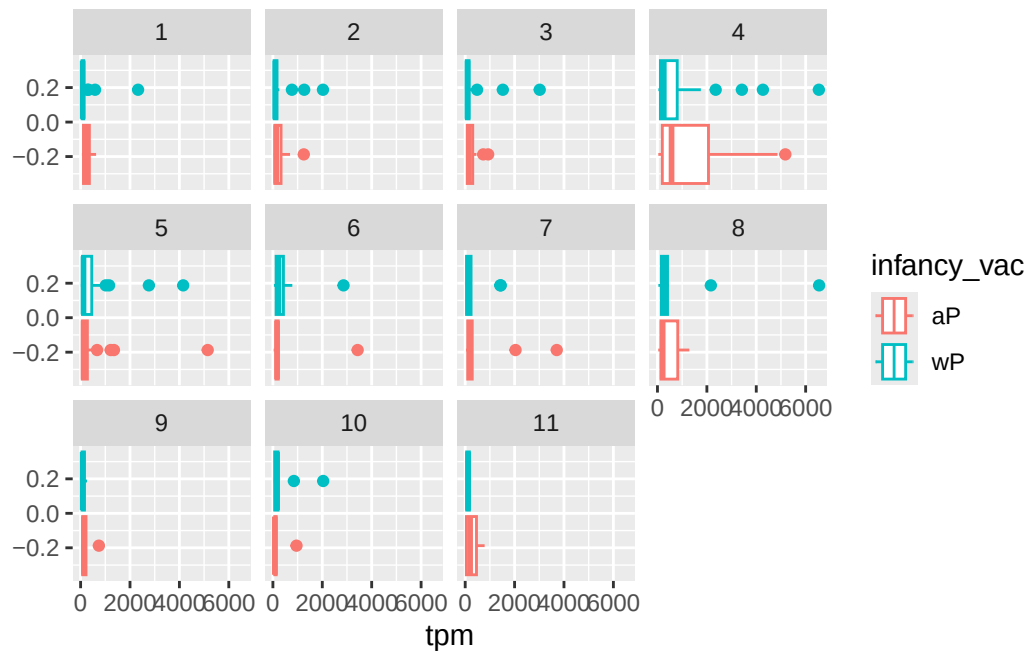
```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



Q. What do you notice about the expression of this gene (i.e. when is it at its maximum level)?

Q. Does this pattern in time match the trend of antibody titer data? If not, why not?

```
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```



```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```